



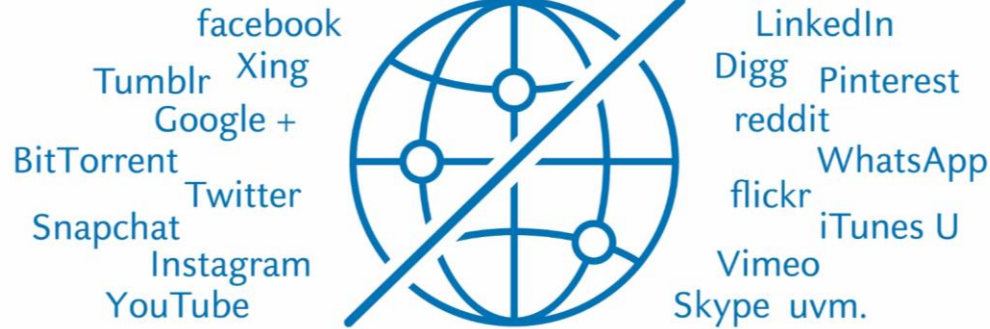
MoPoS – Makromolekulare Chemie II

SoSe 2023



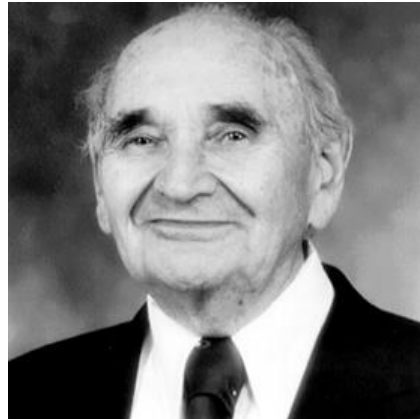
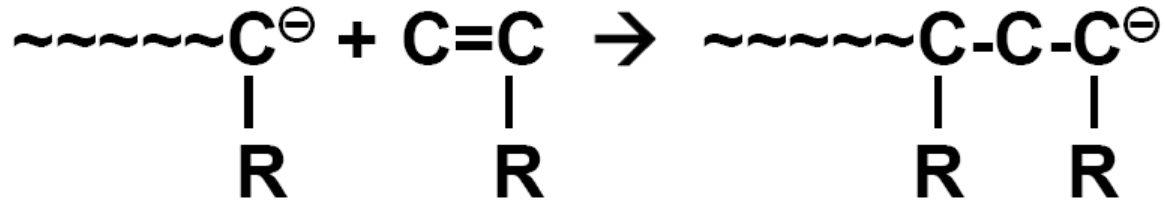
Bitte verzichten Sie in dieser Lehrveranstaltung
auf Bild- und Tonaufzeichnungen.

Weitere Informationen finden Sie im Studierendenportal



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ist nicht gestattet.

Weitere Informationen finden Sie im Studierendenportal



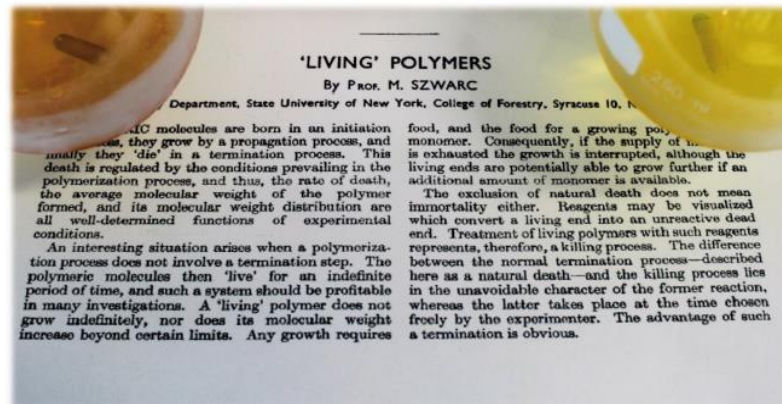
M. Szwarc, 1956:

„Living Polymers“

Nature **178**, 1168 (1956)



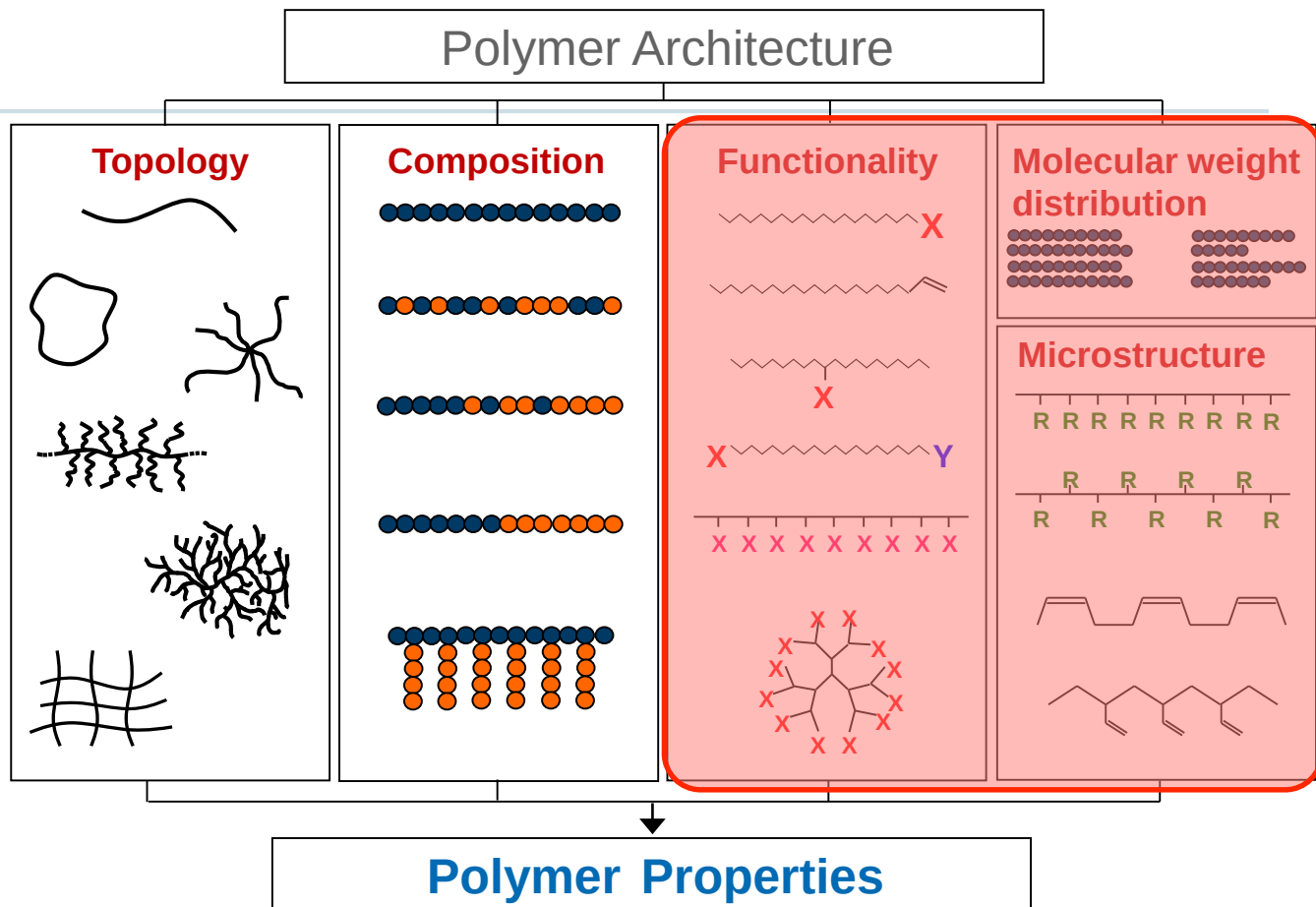
Michael Szwarc



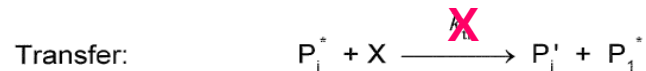
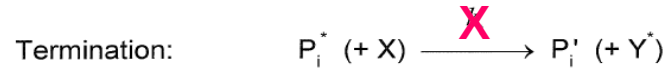
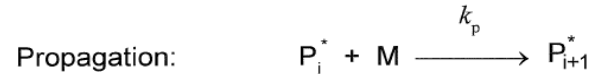
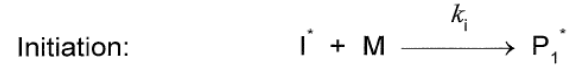
Quicklebig seit 60 Jahren

Die lebende anionische Polymerisation

DANIEL LEIBIG | JAN MORSBACH | EDUARD GRUNE | JANA HERZBERGER |
AXEL H.E. MÜLLER | HOLGER FREY



Basic Principles of Living Polymerization



When termination and transfer are absent

$$(R_t, R_{tr} \ll R_p)$$

→ LIVING POLYMERIZATION

Conditions for a Living Polymerization (1)

- **1. No Termination:** $k_t = 0$ ($R_t \ll R_p$)
- The number of *active* chain-ends, $[P^*]$, is constant:
- Evidence:
- First-order **kinetics** with respect to monomer concentration:

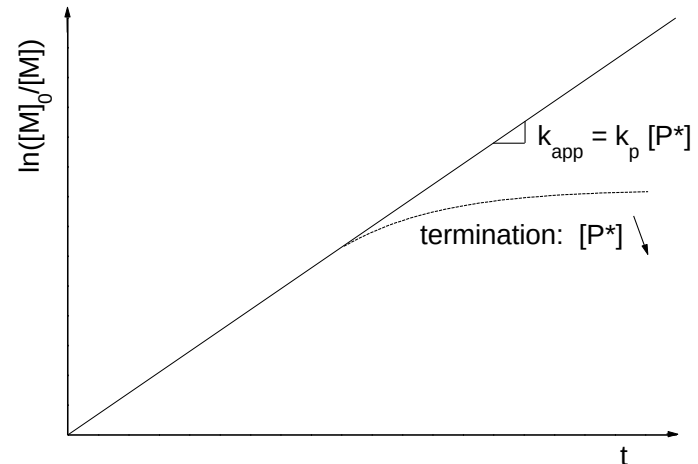
The rate of polymerization is given by

$$R_p = -\frac{d[M]}{dt} = k_p [M][P^*]$$

Integration renders

$$\ln \frac{[M]_0}{[M]} = -\ln(1 - x_p) = k_p [P^*] t = k_{app} t$$

x_p = monomer conversion



Conditions for a Living Polymerization (2)

2. No Transfer: $k_{tr} = 0$ ($R_{tr} \ll R_p$)

→ There are no new chains formed,

→ the total number of polymer chains, $[P]$, (*including the terminated ones!*)

is constant: $[P] = [P^*] + [P_{\text{dead}}] = [I]_0$

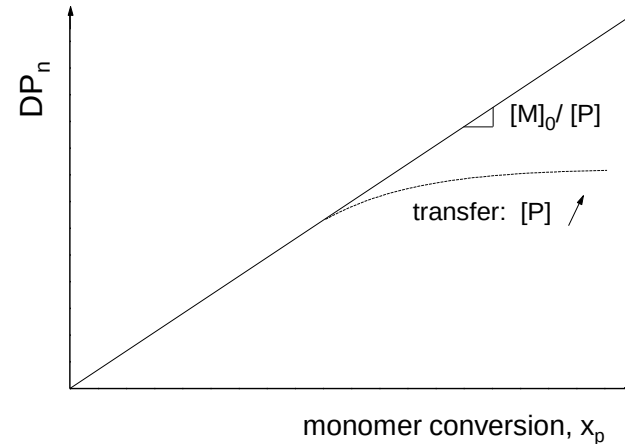
Evidence:

the number-average degree of polymerization, DP_n ,

is a linear function of monomer conversion, x_p :

$$\begin{aligned}\overline{DP}_n &= \frac{\text{number of polymerized monomers}}{\text{number of polymer chains}} \\ &= \frac{[M]_0 - [M]_t}{[P]} = \frac{[M]_0 x_p}{[P]} = \frac{[M]_0 x_p}{f[I]_0}\end{aligned}$$

f = initiator efficiency, $f \leq 1$



Factors leading to broader MWD's

Non-living Processes:

- termination $\rightarrow M_w/M_n \leq 2$ (important for anionic polymerization of (meth)acrylates)
- transfer $\rightarrow M_w/M_n \leq 2$ (important in cationic polymerization of isobutylene)

Living Processes:

- inadequate mixing:

$$t_{\text{mix}} > t_{1/2} \rightarrow M_w/M_n \text{ can be very high}$$

- slow initiation:

$$k_i < k_p \rightarrow M_w/M_n \approx 1.33$$

- reversible polymerization ("scrambling"):

$$\rightarrow M_w/M_n \leq 2$$

- slow equilibria between species of different activity:

$$R_{\text{ex}} < R_p \rightarrow M_w/M_n \text{ can be very high (important for many "new" living/controlled processes)}$$

is a synthetic method to obtain polymers where:

- ♦ the number-average degree of polymerization is given by ratio of monomer over initiator:

$$\overline{DP}_n = \frac{D[M]}{[I]_0} = \frac{[M]_0 \cdot x_p}{[I]_0}$$

- ♦ MWD is narrow: $\overline{M}_w / \overline{M}_n < 1.1$ or 1.2
- ♦ all chains can be functionalized: $P_i^* + X \rightarrow P_i-X$

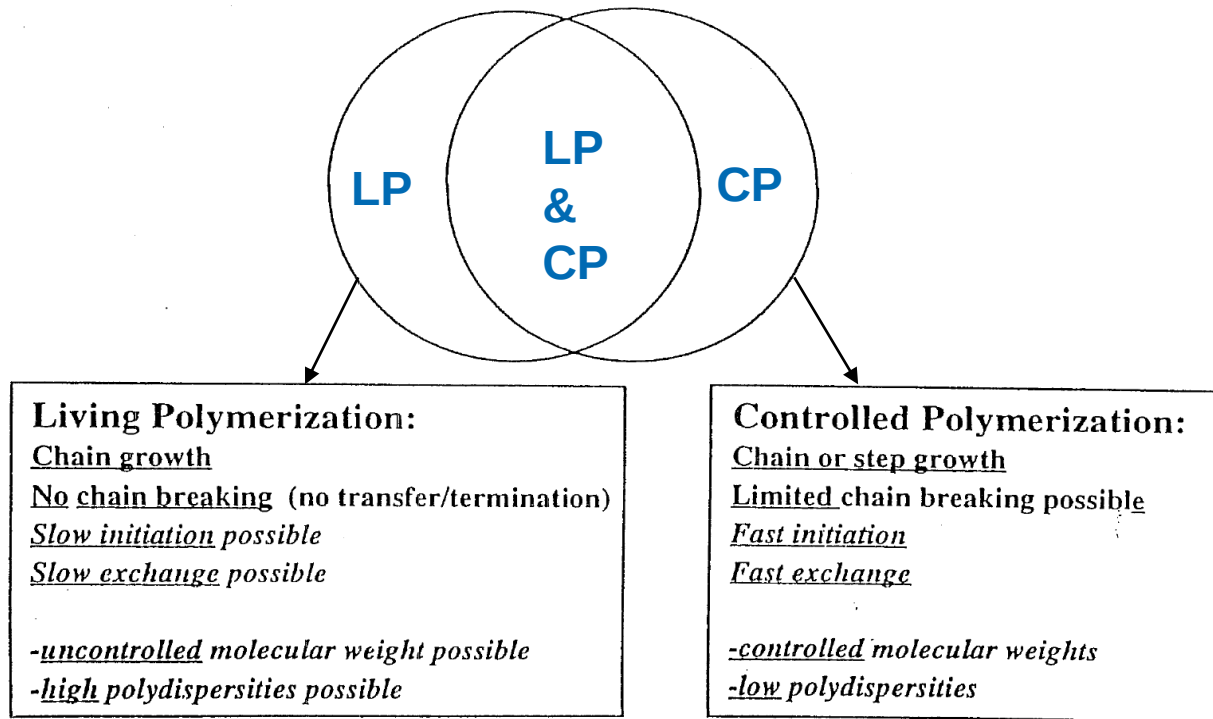
This is obtained if:

- ♦ the rate of initiation and rate of exchange reactions between various active species are fast compared to propagation ($R_{ex}, R_i > R_p$).
- ♦ chain-breaking reactions (termination or transfer) are not too strong ($R_t, R_{tr} \ll R_p$).
Thus a *controlled* polymerization is not necessarily *living*
(e.g. controlled/"living" radical polymerization)

and a *living* polymerization is not necessarily *controlled*
(e.g. slow initiation, slow equilibrium)

Optimum: Living and controlled polymerization

Living versus Controlled



Die wachsenden Kettenenden bei der anionischen Polymerisation sind **Carbanionen**. Der Startschritt besteht entweder in der Addition der Base an das Monomer:

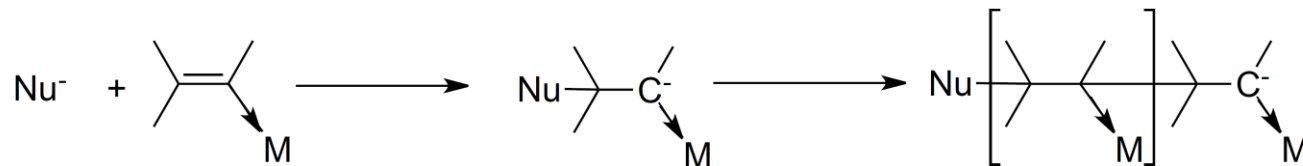


oder in einem Ein-Elektron-Übergang auf das Monomere unter Bildung eines Anionen-Radikals.

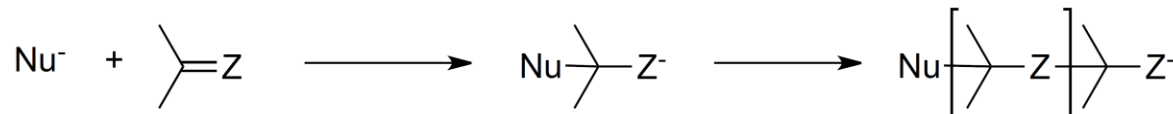


=> Bedingung ist Stabilisierung des Carbanions am Kettenende

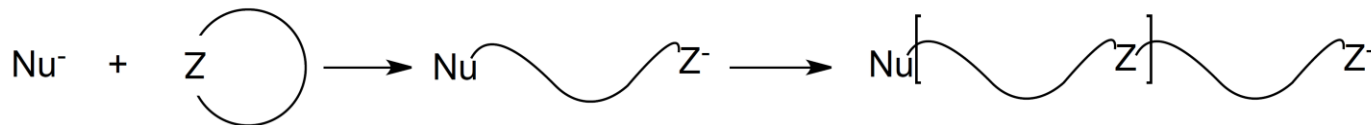
Vinylische
Monomere



Heteroatom-
olefine

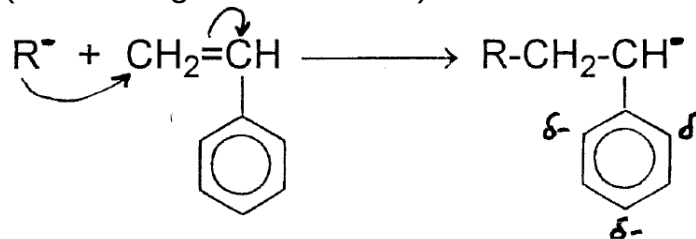


Zyklische
Monomere



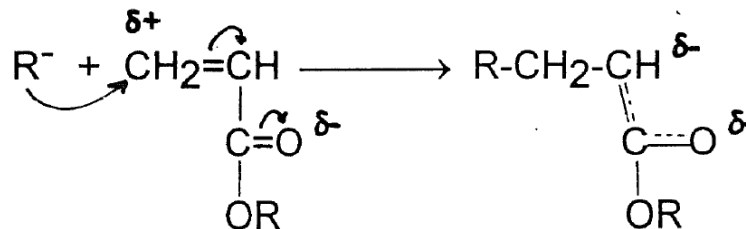
1. Non-polar vinyl compounds (with strong delocalization)

- styrene, α -methyl styrene
- o-, m-, p-alkyl styrenes
- vinyl naphthalene
- dienes (butadiene, isoprene, ...)



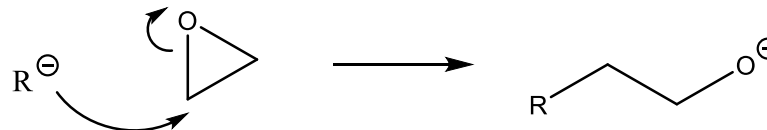
2. Polar electrophilic vinyl compounds (with electron-withdrawing substituents)

- vinyl pyridine
- (meth)acrylates
- N,N-dialkyl acrylamides
- (meth)acrylonitrile

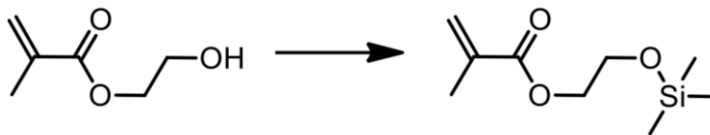


3. Cyclic Ethers, Esters, Siloxanes, ...

→ Ring-Opening Polymerization

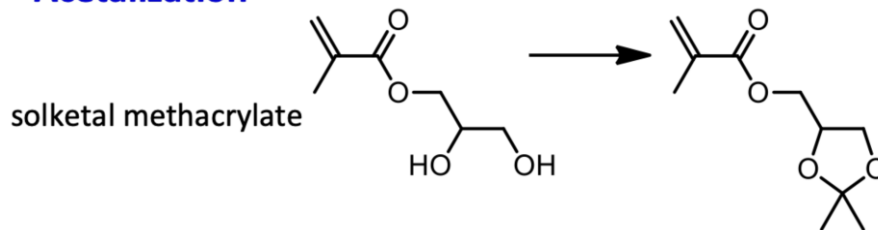


Silylation



2-hydroxyethyl methacrylate (HEMA)

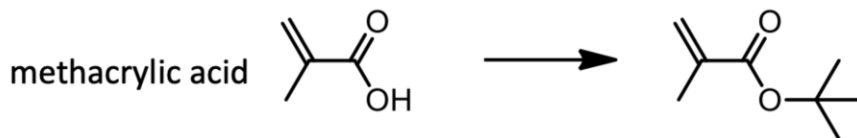
Acetalization



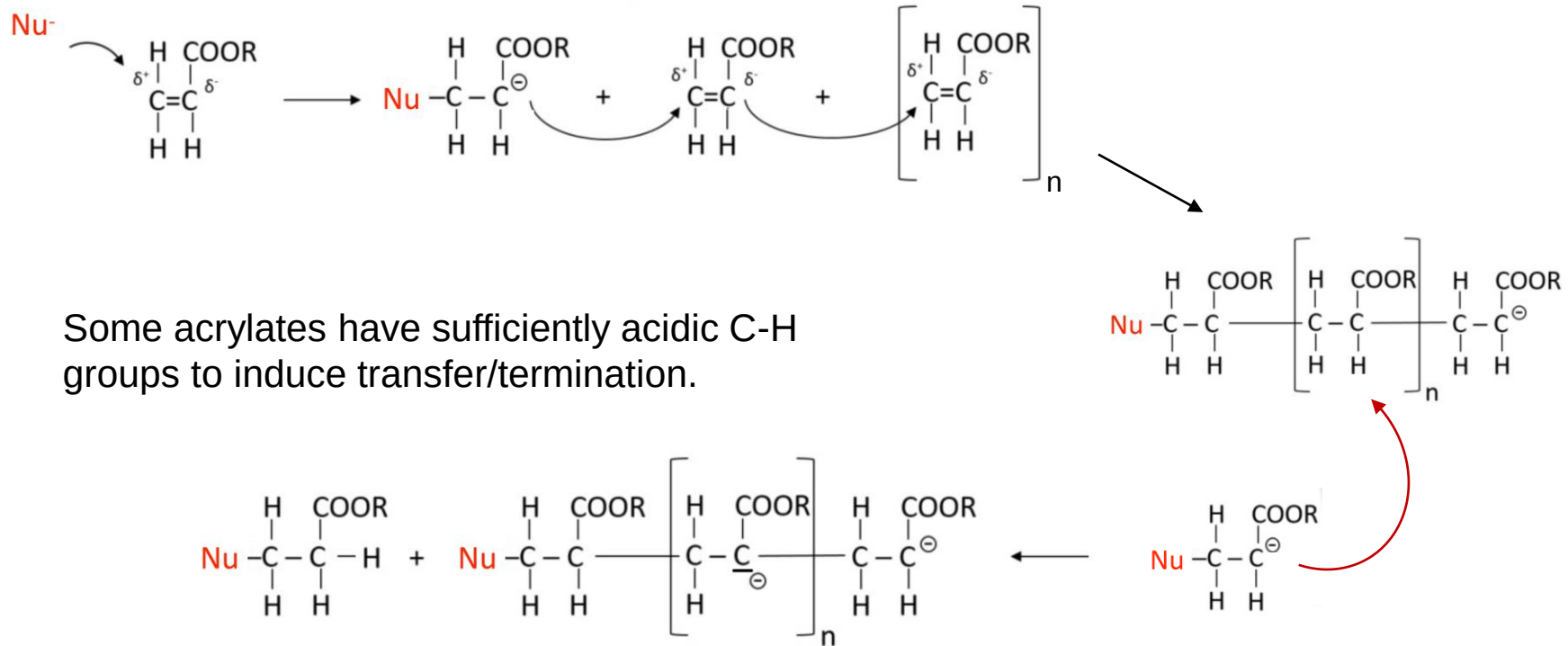
Drawback:

Limited selection of
functional monomers

Esterification



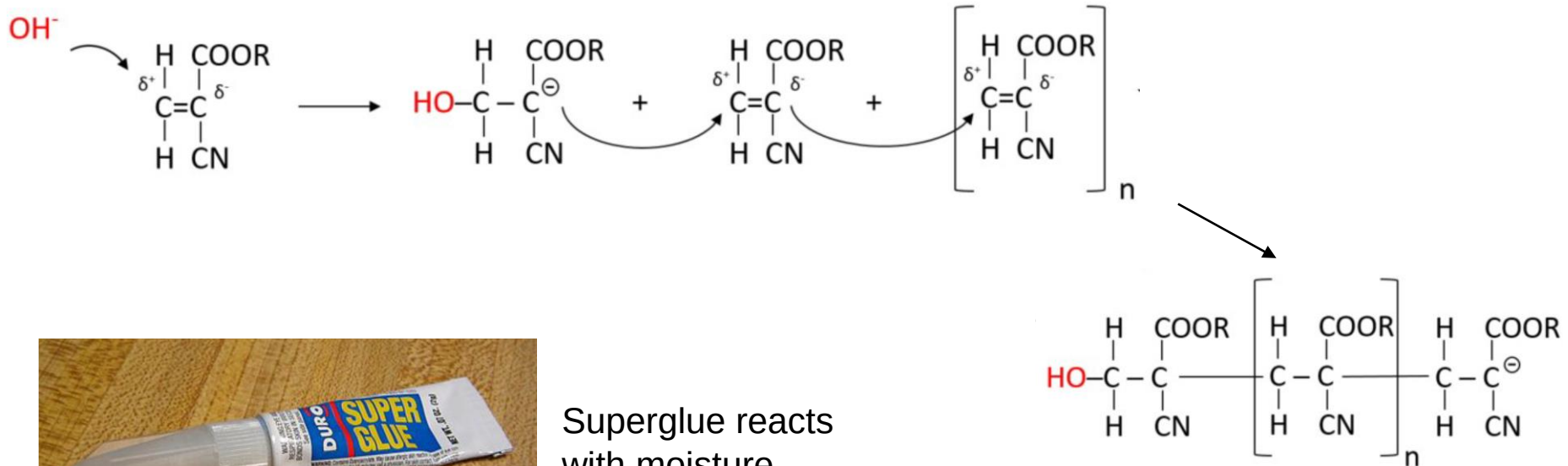
Monomers with Acidic Side Chains



Some acrylates have sufficiently acidic C-H groups to induce transfer/termination.

Highly electron deficient Monomers

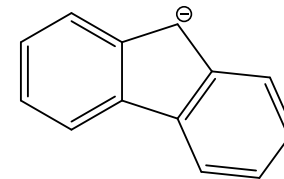
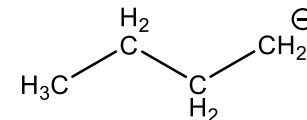
Cyanoacrylates are very reactive, such that water can initiate them..



Superglue reacts with moisture.

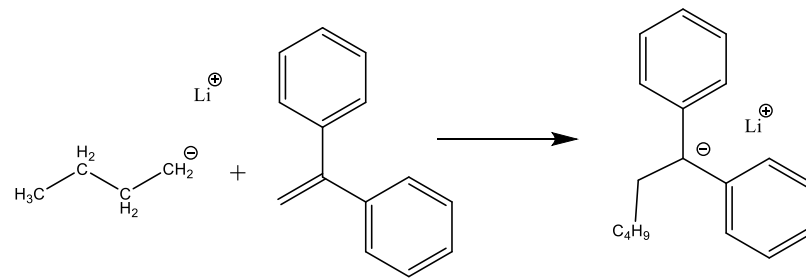
1. Organometallic bases (monofunctional):

- organolithium: n-BuLi, s-BuLi, t-BuLi
- alkali salts of aromatic hydrocarbons: benzyl-Na, cumyl-K, fluorenyl-Na ...
- Grignard reagents (R-MgBr)
- alcoholates (t-BuO-Li, t-BuO-K, ...)
- amides (R₂N-Li)
- (ester) enolates



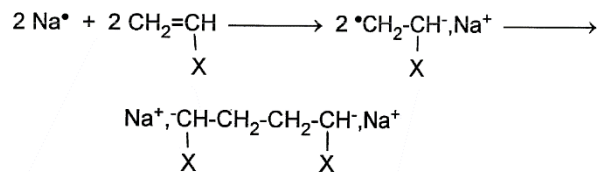
■ Examples

- Alcoholates for cyclic ethers (oxiran)
- s-BuLi for styrene but not methacrylates
- Better cuprates (2LiR + CuX → Li⁺ (R₂Cu)⁻ + LiX) or transfer to diphenylethylene



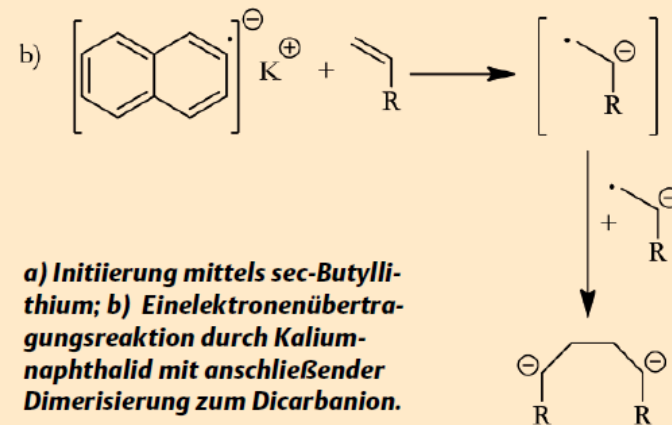
2. Electron transfer agents (bifunctional):

- alkali metals (heterogeneous!):



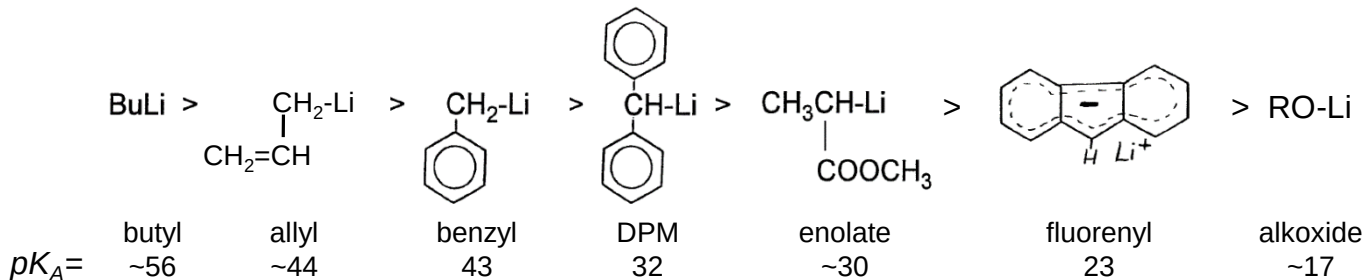
(W. Schlenk 1914; K. Ziegler 1929)
→ Buna = Butadien + Natrium

- Naphthalene acts as soluble electron transfer agent to styrene
- Works in polar solvents like THF
- Developed by Szwarc and enables **homogeneous** anionic polymerization



The reactivity of an initiator depends on:

- the nucleophilicity of the anion,
(roughly correlates with the pK_A value of the non-metalated compound):



- the ionic radius of the counterion: $NR_4^+ > Cs^+ > K^+ > Na^+ > Li^+$
- the polarity of the solvent: THF > toluene

The nucleophilicity of the initiator should be equal to or higher than the electrophilicity of the monomer (pK_A of the hydrogenated monomer)

- Due to the high nucleophilicity of the initiators (and propagating chain ends) it is absolutely necessary to avoid oxygen, water and protic impurities:
- This implies
 - - **aprotic solvents**:
polar: THF, ...
non-polar: toluene, cyclohexane, hexane, ...
 - - rigorous **purification of reagents**
 - - handling of reagents in **vacuum** or under **inert gas**.
- Due to the absence of termination, the concentration of active species can be much higher than in radical polymerization. Thus, the **rates** sometimes can be **very high**, (e.g., $t_{1/2} < 1$ s for MMA in THF at room temperature).
- In order to control the polymerization it may be necessary to
 - - use specially designed reactors (fast mixing: flow tube)
 - - add monomer slowly (vapour phase)
 - - work at low temperatures (-78 °C)



Abb. 5 Polymerisationslösungen der lebenden anionischen Polymerisationen von 2-Vinylpyridin (dunkelrot, links), Styrol (orange, mittig) und Isopren (gelb, rechts) in Tetrahydrofuran.

High Vacuum (Break-Seal) Technique

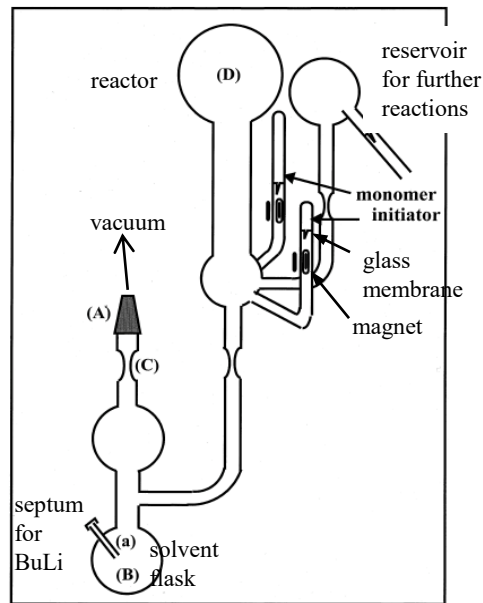


Figure 9. Apparatus for the preparation of linear homo- and copolymers, along with symmetric star homopolymers and star-block copolymers, consisting of the purge section and the main polymerization reactor.

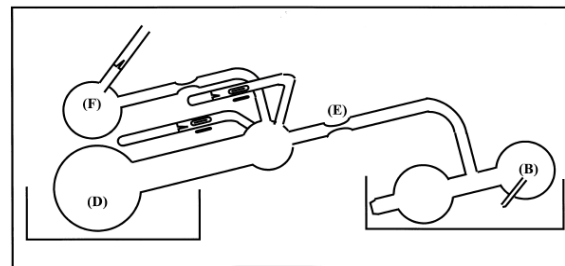


Figure 10. Intermediate step involving rinsing of the polymerization apparatus by distillation of the solvent from the purge section to the main apparatus.

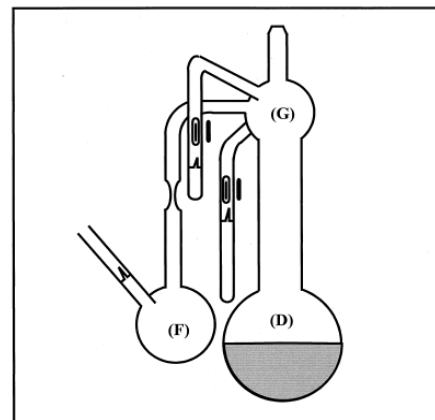
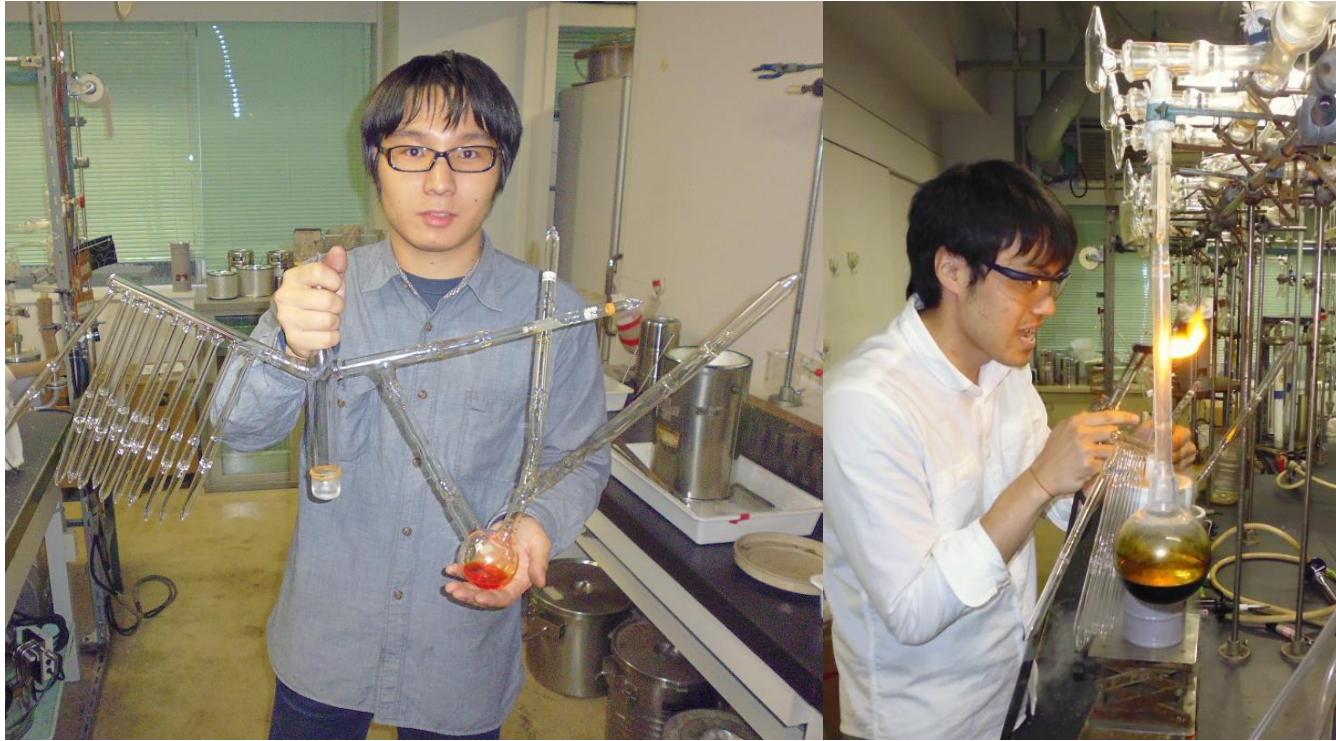
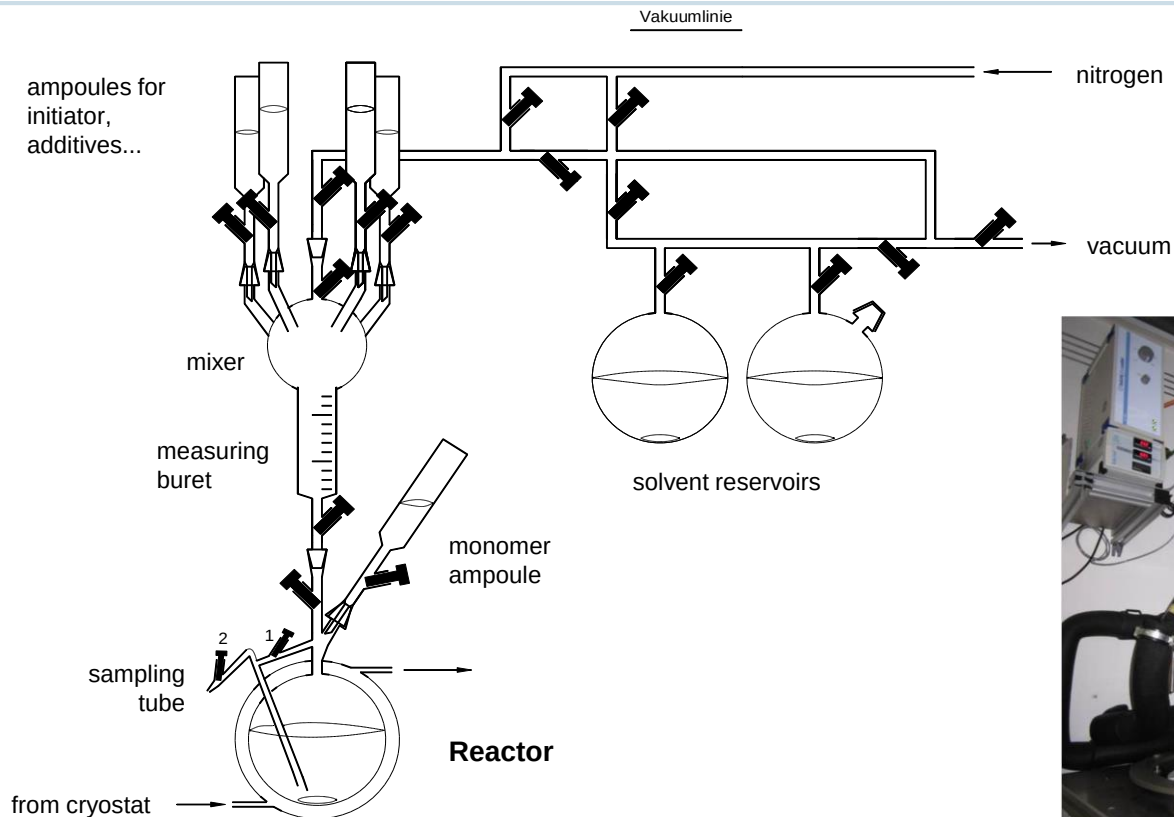


Figure 11. The main polymerization apparatus.

Group of Prof. Ishizone (Japan)



Laboratory Reactor for Anionic Polymerization

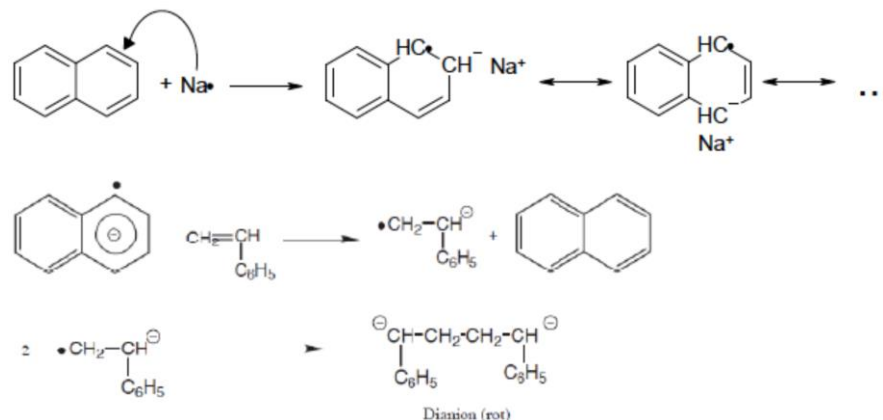
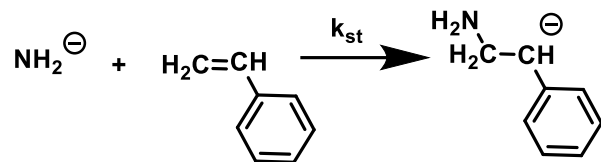


Schlenk Line for Anionic Polymerization



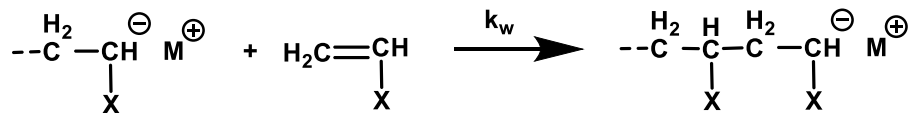
1. Initiation / Start of the polymerization

- Either through addition of base or electron transfer.



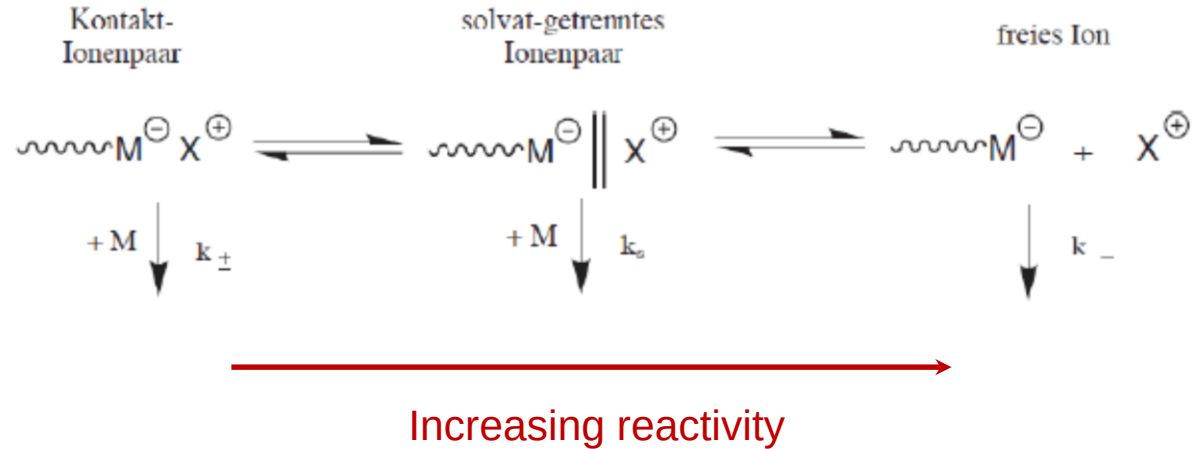
2. Chain Growth

Monomer addition to carbanion, reaction speed determined by type of metal/ion pair



Influence of Solvent on Reaction speed

Reactivity of
growing chain end

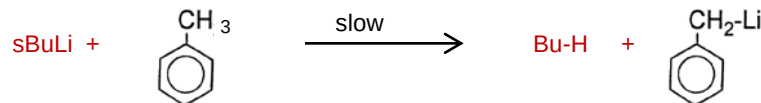


- At high temperatures mostly contact ion pairs (entropy of solvent molecules)
- At low temperatures mostly solvated ion pairs

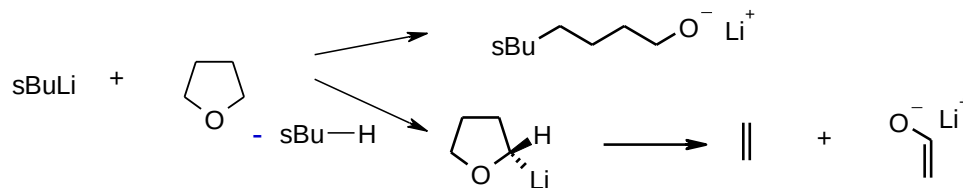
Side Reactions in Anionic Polymerization

3. Termination and transfer reactions

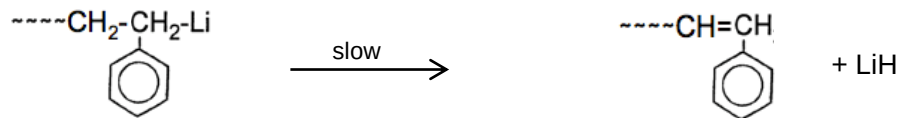
Transfer to Solvent (Toluene):



Termination by Reaction with THF:



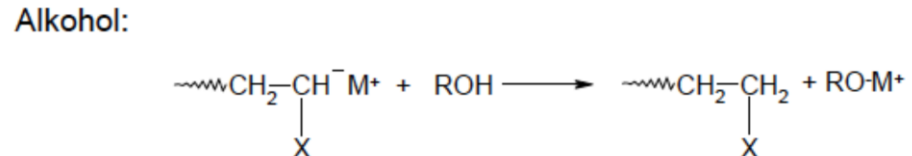
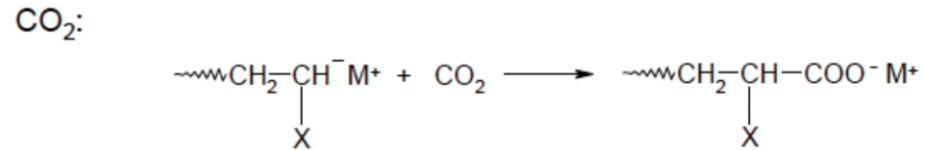
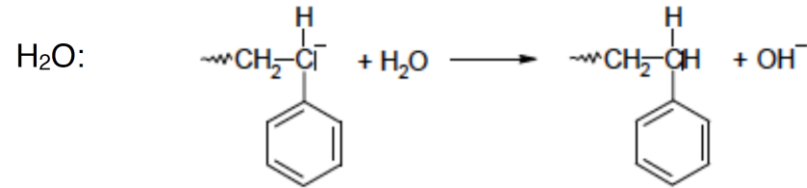
Termination by Hydride Elimination:



Side Reactions in Anionic Polymerization

3. Termination and transfer reactions

Chain termination normally only occurs with impurities, but can be used to intentionally install specific **end-groups**.

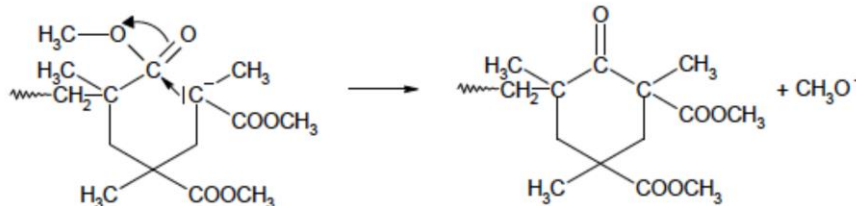
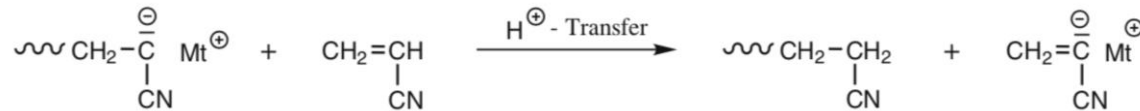
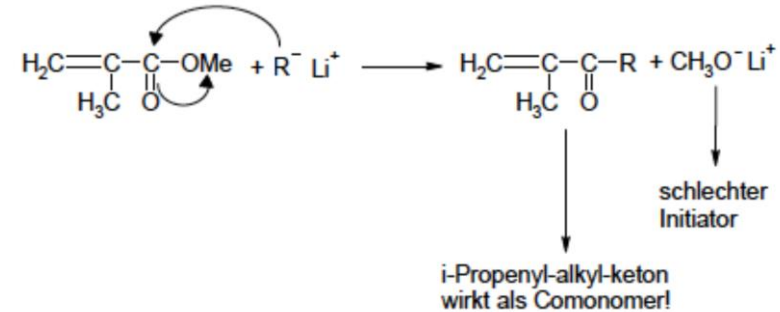


Side Reactions in Anionic Polymerization

3. Termination and transfer reactions

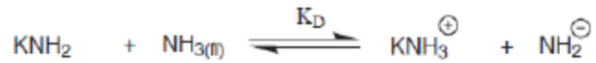
Other side reactions

- Attack of the wrong monomer position (carbonyl groups)
- Abstraction of acidic protons
- Intramolecular backbiting

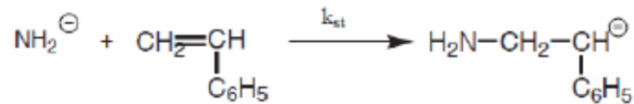


Kinetics of AP – with Chain Transfer

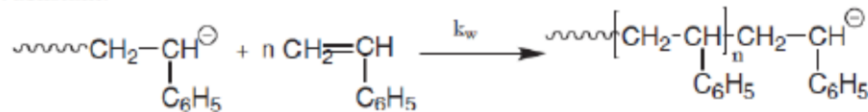
Styrene with KNH₂ in liquid ammonia



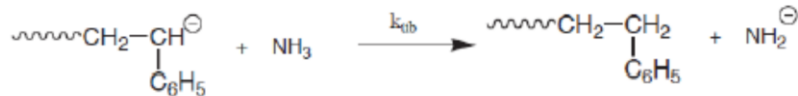
Start:



Wachstum:



Übertragung (mit dem Lösungsmittel):



Steady state: $v_{st} = v_{tr}$

$$k_{tr} \cdot [\text{NH}_3] \cdot \sum_i [\text{M}_i^-] = k_{st} \cdot \sqrt{K_D \cdot [\text{KNH}_2]} \cdot [\text{M}] \Rightarrow \sum_i [\text{M}_i^-] = \frac{k_{st} \cdot \sqrt{K_D \cdot [\text{KNH}_2]} \cdot [\text{M}]}{k_{tr} \cdot [\text{NH}_3]}$$

$$K_D = \frac{[\text{K}^+][\text{NH}_2^-]}{[\text{KNH}_2]} = \frac{[\text{NH}_2^-]^2}{[\text{KNH}_2]} \Rightarrow [\text{NH}_2^-] = \sqrt{K_D [\text{KNH}_2]}$$

$$v_{\text{start}} = \frac{d[\text{M}^-]}{dt} = k_{st} [\text{NH}_2^-] [\text{M}] = k_{st} \sqrt{K_D [\text{KNH}_2]} [\text{M}]$$

$$v_w = -\frac{d[\text{M}]}{dt} = k_w \cdot [\text{M}] \cdot \sum_i [\text{M}_i^-]$$

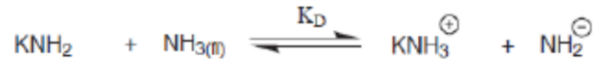
$$v_{tr} = k_{tr} \cdot [\text{NH}_3] \cdot \sum_i [\text{M}_i^-]$$

$$\begin{aligned} v_w = -\frac{d[\text{M}]}{dt} &= k_w \cdot [\text{M}] \cdot \sum_i [\text{M}_i^-] \\ &= \frac{k_w \cdot k_{st}}{k_{tr}} \cdot \sqrt{K_D} \cdot \frac{\sqrt{[\text{KNH}_2]}}{[\text{NH}_3]} [\text{M}]^2 \end{aligned}$$

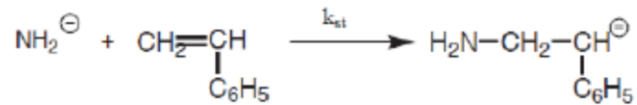
Kinetik 2. Ordnung

Kinetics of AP – with Chain Transfer

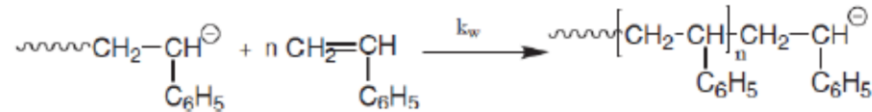
■ Styrene with KNH₂ in liquid ammonia



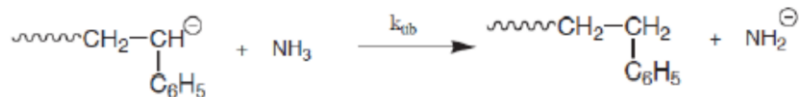
Start:



Wachstum:



Übertragung (mit dem Lösungsmittel):



Degree of Polymerization / kinetic chain length

$$X_n = \nu = \frac{v_w}{v_{\text{start}}} = \frac{k_w \cdot k_{\text{st}} \cdot \sqrt{K_D} \cdot \frac{\sqrt{[\text{KNH}_2}]}{[\text{NH}_3]} [M]^2}{k_{\text{tr}} \cdot \sqrt{K_D} \cdot [\text{KNH}_2] \cdot [M]}$$

$$X_n = \frac{k_w}{k_{\text{tr}}} \cdot \frac{[M]}{[\text{NH}_3]}$$

Depends on
Monomer and
solvent
concentration

Anionic Polymerization without Transfer/Termination

- The most important conclusions on the mechanism have been drawn from kinetic measurements of the polymerization of non-polar monomers, e.g. **styrene**.
- (*M. Szwarc, Syracuse; G.V. Schulz, Mainz; 1964-70*)

- Rate of polymerization:
$$R_p = - \frac{d[M]}{dt} = k_p [M] [P^*],$$

- Concentration of active chain-ends: $[P^*] = \text{const.} = [I]_0$

- Integration leads to the pseudo first-order relation:

$$\ln \frac{[M]_0}{[M]} = k_p [P^*] t = k_{\text{app}} t.$$

k_p , rate constant of propagation
 k_{app} , apparent rate constant (slope of first-order plot)

Anionic Polymerization without Transfer/Termination

$$v_w = -\frac{d[M]}{dt} = k_w \cdot [M] \cdot \sum_i [M_i^-] = k_w \cdot [M] \cdot [I]_0$$

$$\Rightarrow -\frac{d[M]}{[M]} = k_w \cdot [I]_0 \cdot dt \Rightarrow -\int_{[M]_0}^{[M](t)} \frac{d[M]}{[M]} = k_w \cdot [I]_0 \cdot \int_0^t dt$$

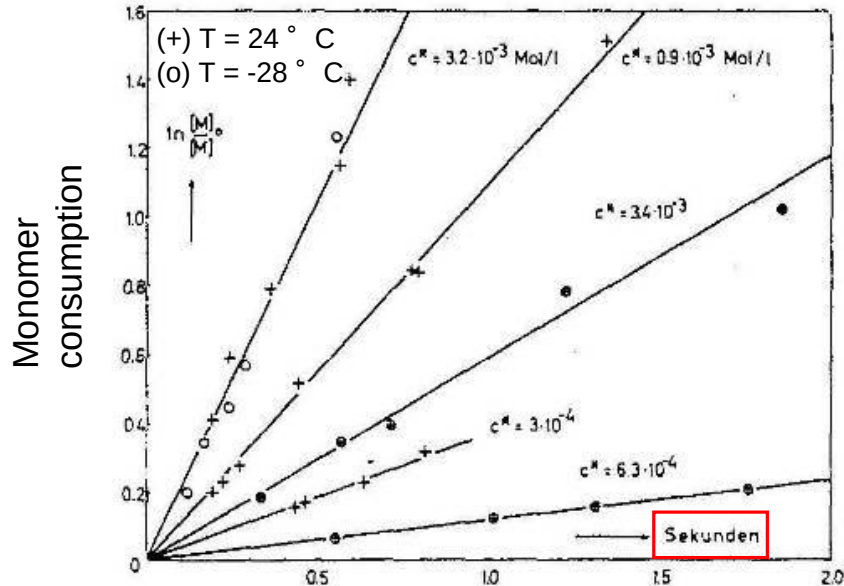
$$\Rightarrow -\ln[M] \Big|_{[M]_0}^{[M](t)} = k_w \cdot [I]_0 \cdot t \Big|_0^t$$

$$\Rightarrow -\ln[M](t) + \ln[M]_0 = k_w \cdot [I]_0 \cdot t$$

$$\Rightarrow \ln \frac{[M]_0}{[M](t)} = k_w \cdot [I]_0 \cdot t = k_{app} \cdot t$$

First-Order Plot of Kinetics

Polymerization of styrene with Na⁺ counterion in THF



$$\ln \frac{[M]_o}{[M](t)} = k_w \cdot [I]_o \cdot t = k_{app} \cdot t$$

$$k_{app} = k_p[P^*]$$

is the slope of the first-order plot

Linear plot (constant slope)

→ [P*] = const.

→ **no termination**

→ **fast initiation**



Molecular Weight vs. Conversion

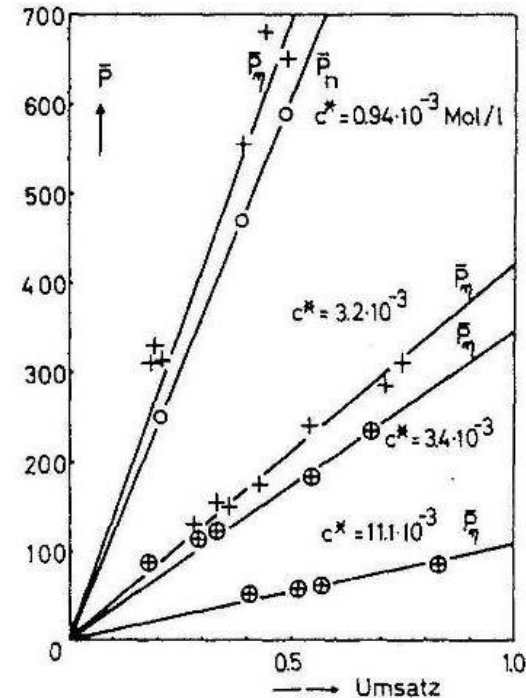
- Linear plot (constant slope)
- the number of all chains, $[P]$, is constant

- no transfer
- fast initiation

- $M_w/M_n < 1.02$

$$\Rightarrow X_n = p \cdot \frac{[M]_0}{[I]_0}$$

MW can be pre-determined



A Constant is a Constant is...???

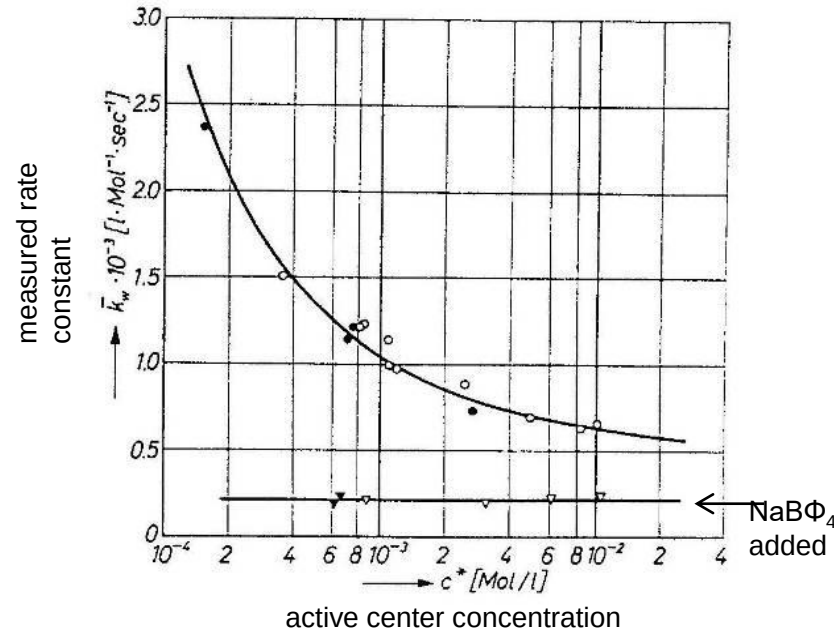
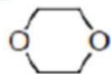


Abb. 4. Mittlere Geschwindigkeitskonstante $\bar{k}_w$ in Abhängigkeit von der Konzentration der aktiven Zentren. ○ ● bzw. △ ▲ ohne bzw. mit Zusatz von Na-Tetraphenylborat. Ausgefüllte Punkte: $\bar{P}_n$ durch Fraktionierung bestimmt

Types of Ion Pairs

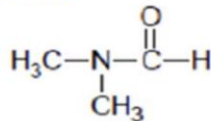
pentane
hexane
cyclohexane
benzene
dioxane



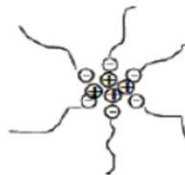
1,2 dimethoxyethane
 $\text{CH}_3\text{OCH}_2\text{CH}_2\text{OCH}_3$
tetrahydrofuran



dimethyl formamide



increasing
polarity



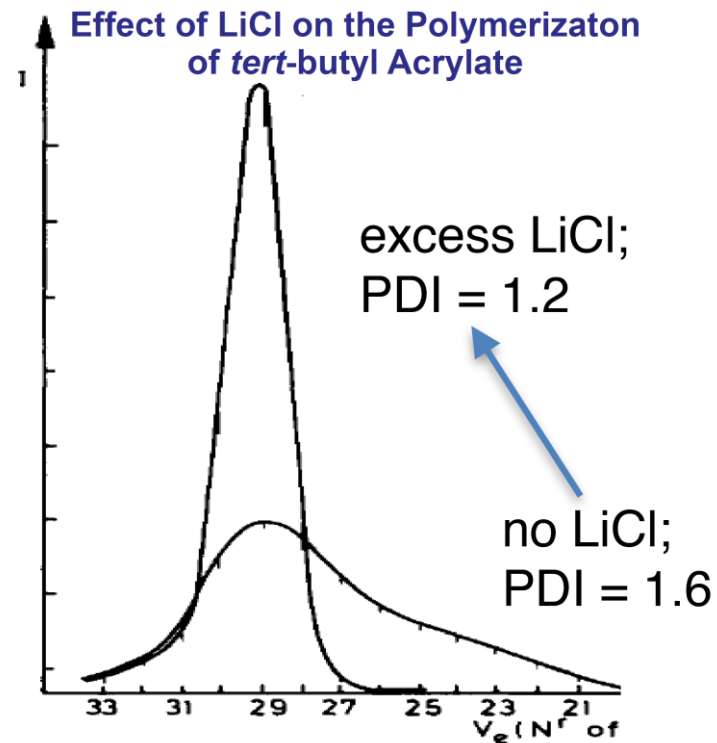
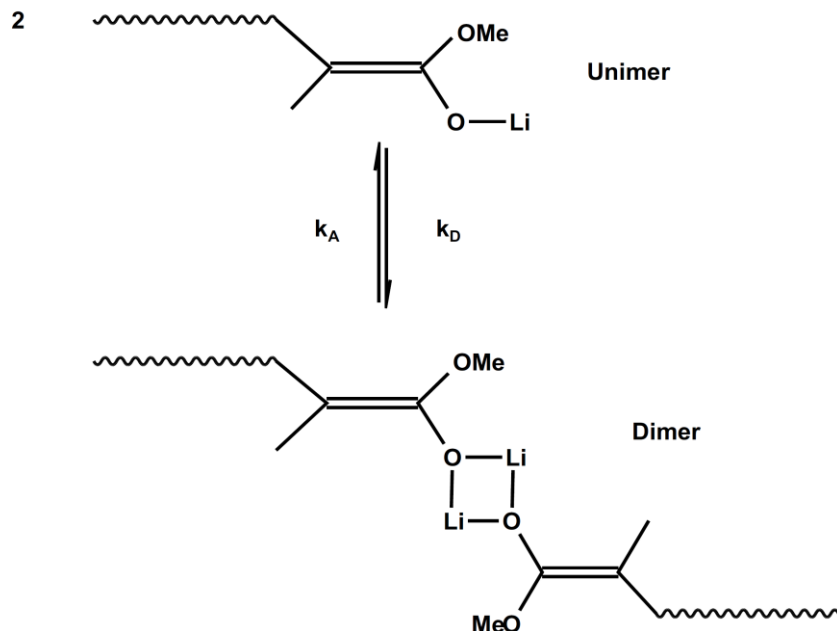
Different monomer addition rates.

Fast exchange between species
needed for narrow MWD

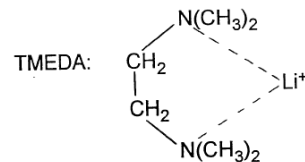
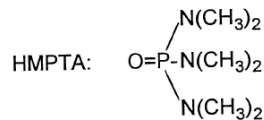
Effects of aggregation - Methacrylates

Enolate-Formation for Methacrylates

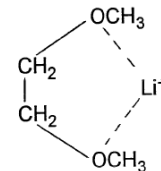
Dimer formation, reduction in polymerization rate



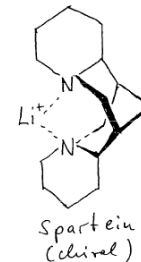
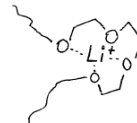
Solvating and Chelating agents (Lewis Bases):



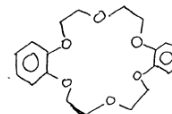
Dimethoxyethane (glyme):



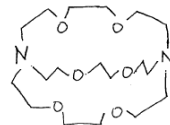
Di-, tri-, tetra-glyme,...
Poly(ethylene oxide)



Crown ethers (Pederson):
(Dibenzo-18-crown-6)



Cryptands (Lehn)
(Cryptand 2.2.2)

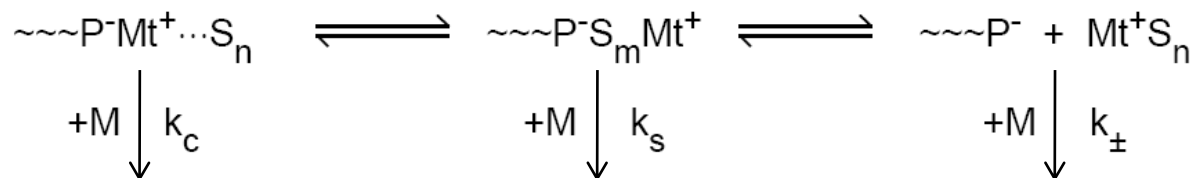


Polar Solvents:

Contact Ion Pair
(peripherally solvated)

Solvent-Separated
Ion Pair

Free Anion



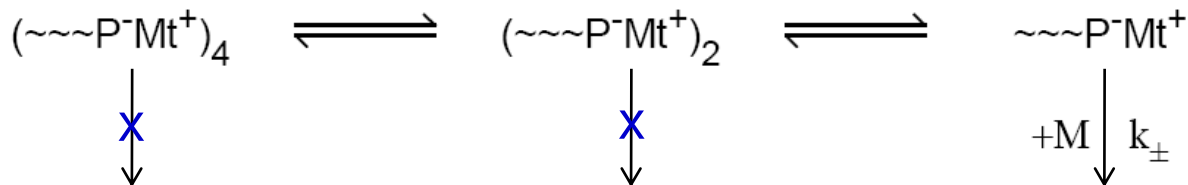
Non-polar Solvents:

chelating agent

Tetrameric
Aggregates

Dimeric
Aggregates

Contact Ion Pair
(non-solvated)



Regio- and Stereochemistry of Dienes:

Control of Microstructure

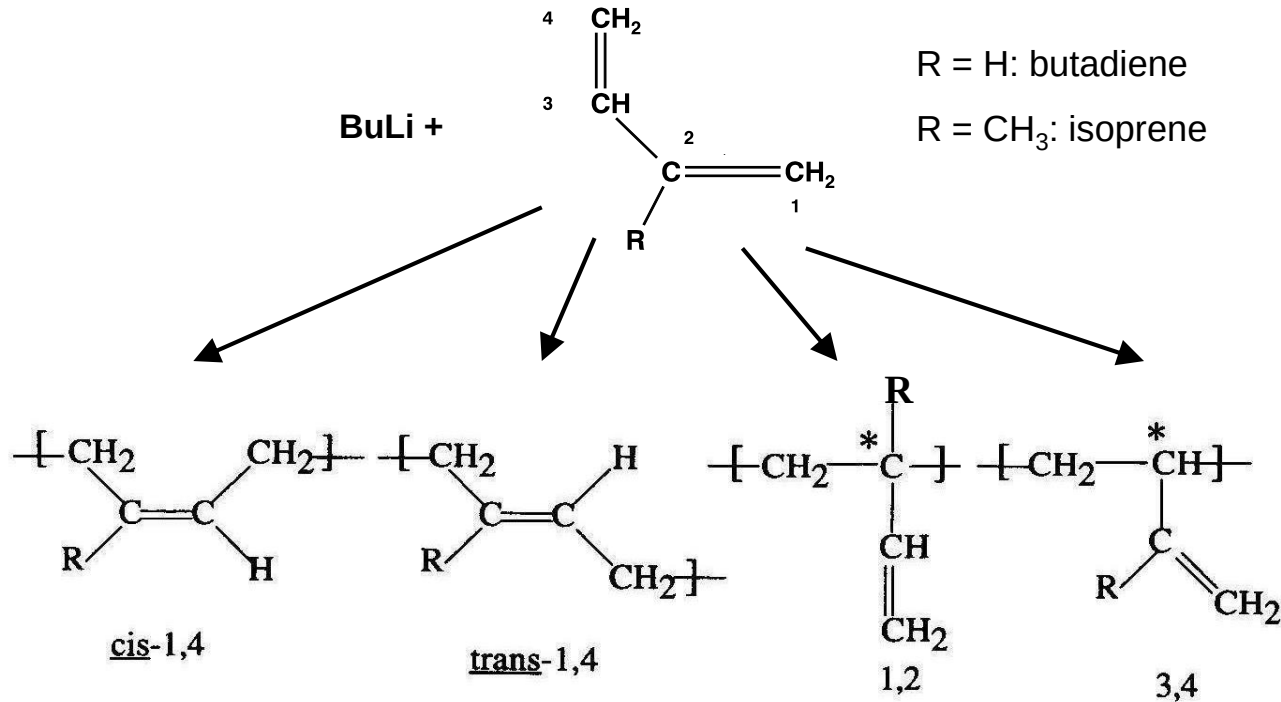


Table 1 Configuration of Polybutadiene and Polyisoprene as a Function of Counterion and Solvent

Polymerization Conditions		Counterion	Temperature (°C)	cis	Microstructure		
Monomer	Solvent				trans	1,2	3,4
Butadiene	hexane	Li ⁺	20 ¹	0.68	0.28	0.04	—
Butadiene	hexane	Li ⁺	20 ²	0.30	0.62	0.08	—
Butadiene	THF	Li ⁺	0	0.06	0.06	0.88	—
Butadiene	THF	Li ⁺	−78	~ 0	0.08	0.92	—
Butadiene	THF	Na ⁺	0	0.06	0.14	0.80	—
Butadiene	THF	Na ⁺	−78	~ 0	0.14	0.86	—
Butadiene	THF	K ⁺	0 or −78	0.05	0.28	0.67	—
Butadiene	diethyl ether	Li ⁺	0	0.08	0.17	0.75	—
Butadiene	diethyl ether	K ⁺	0	0.11	0.34	0.55	—
Isoprene	cyclohexane	Li ⁺	30 ¹	0.94	0.01	—	0.05
Isoprene	cyclohexane	Li ⁺	30 ²	0.76	0.19	—	0.05
Isoprene	cyclohexane	Na ⁺	15	—0.44—	—	0.06	0.50
Isoprene	cyclohexane	K ⁺	15	—0.59—	—	0.05	0.36
Isoprene	cyclohexane	Cs ⁺	15	—0.69—	—	0.04	0.27
Isoprene	THF	Li ⁺	30	—0.12—	—	0.29	0.59
Isoprene	THF	Na ⁺	0	—0.11—	—	0.19	0.70
Isoprene	diethyl ether	Li ⁺	20	—0.35—	—	0.13	0.52
Isoprene	diethyl ether	K ⁺	20	—0.38—	—	0.19	0.43
Isoprene	diethyl ether	Cs ⁺	20	—0.52—	—	0.16	0.32

Ionization suppressant added in the case of THF.

¹ At monomer/initiator ratio 5×10^4 .

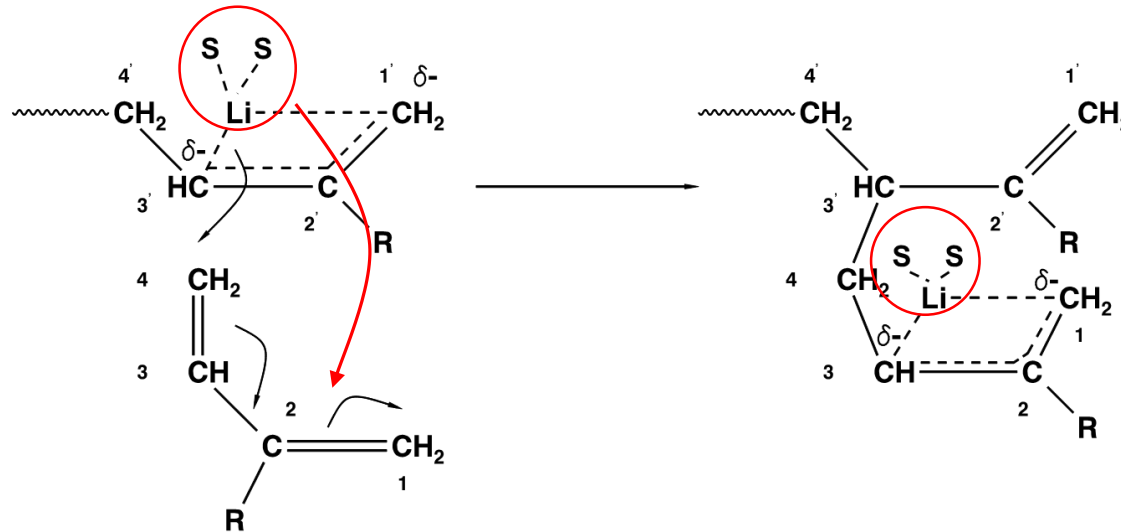
² At monomer/initiator ratio ~ 17.

Conc. of active species, counterions, solvent, temperature, additives, ... determine microstructure

In **polar solvents** the *cis* form is thermodynamically stable.

Charge density at C-3' is higher than at C-1'

→ 1,2- or 3,4-addition is favoured → vinyl-type structure



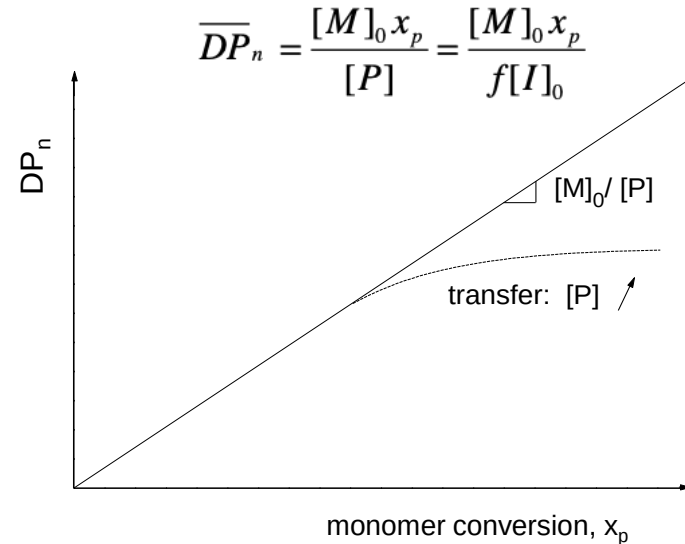
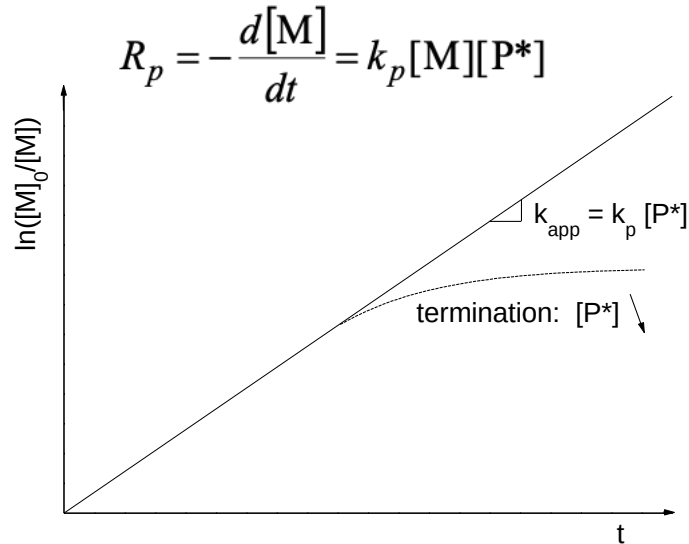
Monomer approaches in *s-cis* fashion → *cis* chain end

Anionische Polymerisation

Zusammenfassung

1. Keine Terminierung: $k_t = 0$ ($R_t \ll R_p$)

2. Keine Übertragungsreaktionen



Polymer Architecture

